

AWS re:Invent

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API 313-R

Understanding design in event-driven architecture

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About me

David Boyne

Senior Developer Advocate – AWS Serverless
Event-driven architectures and learn in public

Father of two

1st – Vegas and re:Invent

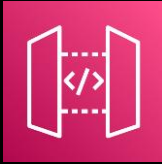


Why are we here today?



Events are the language of serverless applications

Event-driven architectures becoming more accessible



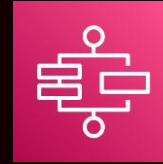
Amazon
API Gateway



Amazon
DynamoDB



Amazon
EventBridge

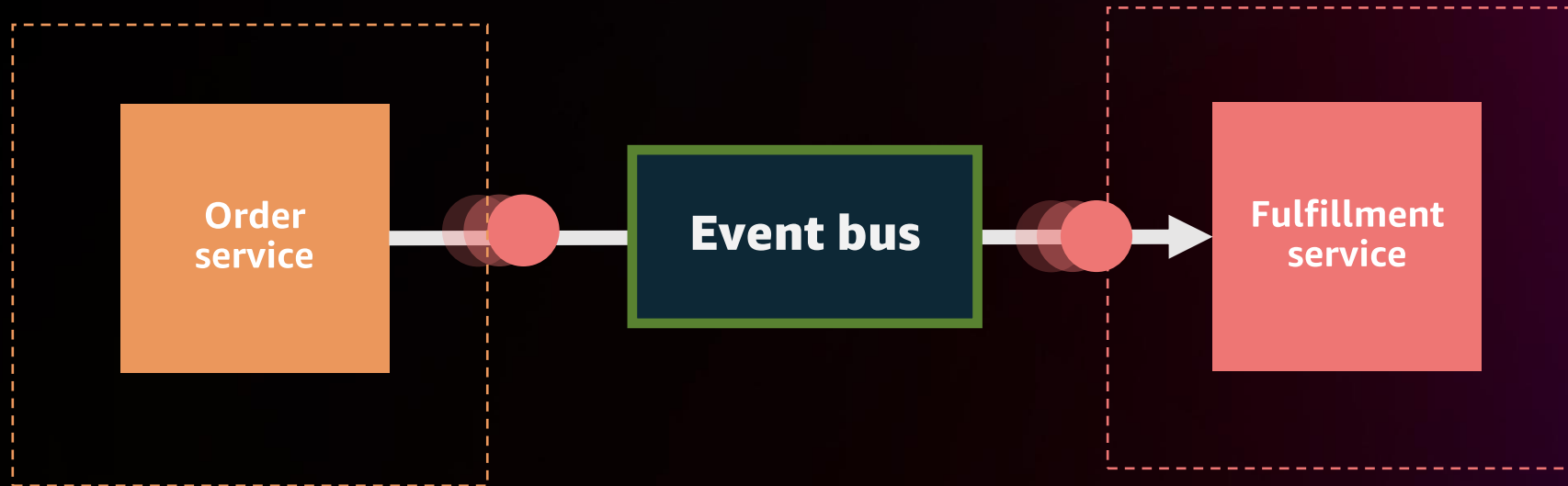


AWS Step
Functions

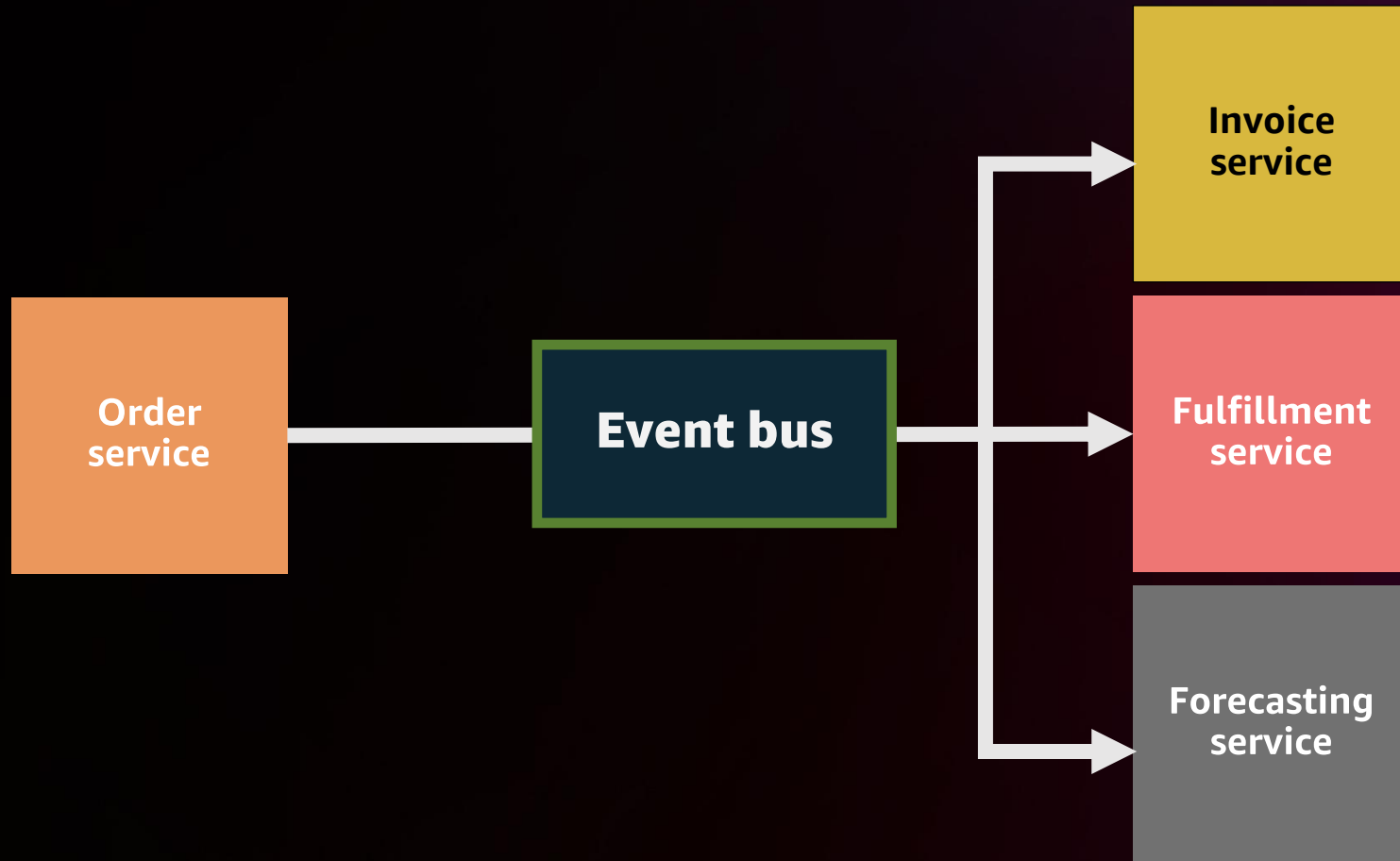


AWS
Lambda

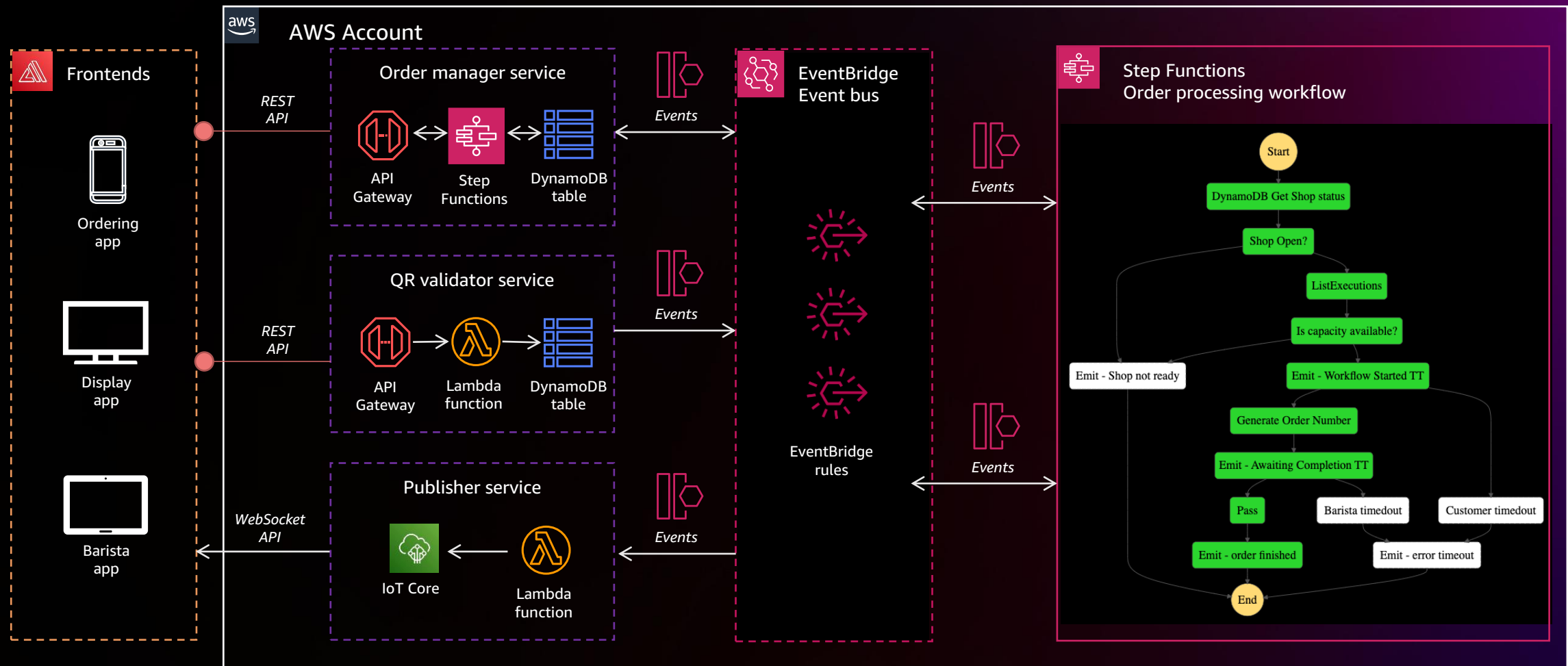
Decoupled architectures



Evolutionary architecture



Event-driven architectures provide agility



serverlesspresso

Common questions start to emerge

- How do we **identify** events?
- How do we **design** our events?
- How can we **scale** event-driven architectures?



**“Producers should not
know about consumers ...”**

How you design your events **impacts your** **architecture**



Agenda

- Identify: Explore and understand
- Design: Structure and standards
- Implement: Libraries and documentation
- Summary

Identify:

Explore and understand



Design: Structure and standards

Implement: Libraries and documentation

Summary

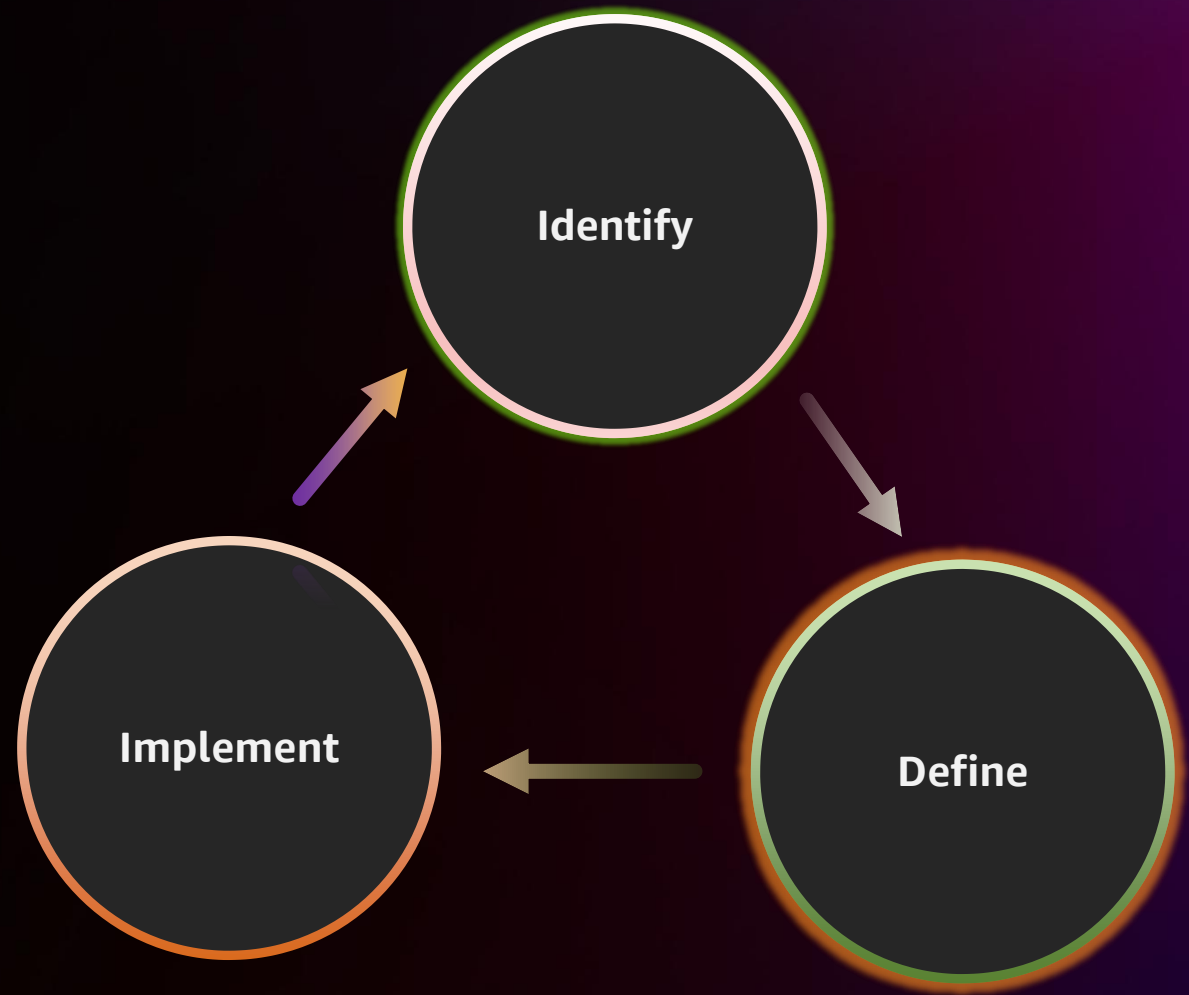


Common questions start to emerge

- How do we **identify** events?
- How do we **design** our events?
- How can we **scale** EDA solutions within our org?

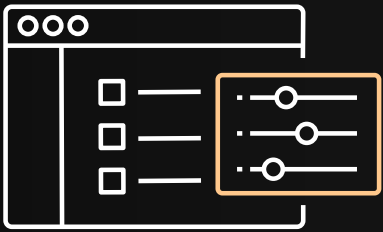


Event-first thinking



Event-first thinking: Identify

Identify



Behavior of system

Understand
system behavior



Event storming

Collaborate with
domain experts



Domain

Identify bounded
context and
domains

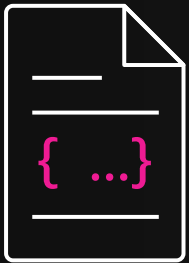


Ubiquitous language

Speak the same
language

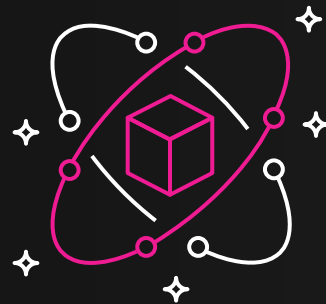
Event-first thinking: Define

Define



Event standards

Define standards
in event payloads



Boundaries

Clear boundaries
between domains



Documentation

Events, producers,
and consumers



Naming conventions

Align to clearly
defined domains

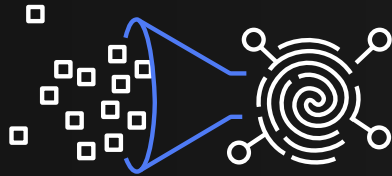
Event-first thinking: Implement

Implement



Libraries

Develop shared
code to help



Event patterns

Consider trade-
offs of design
choices



Domain vs. integration

Internal vs. external
events



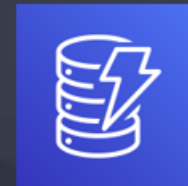
Domain events

Events that the
business
understands

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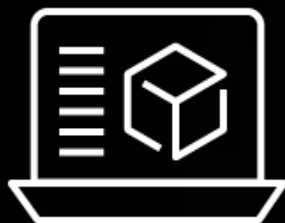
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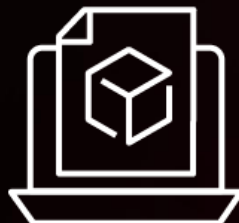
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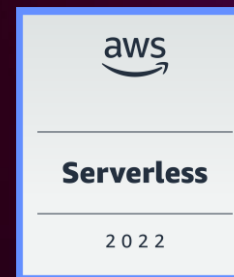
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<https://s12d.com/serverless-learning>



Recommended talks

- API302 – Building global event-driven applications
- API309 – Advanced serverless workflow patterns and best practices
- API307 – Designing event-driven integrations using Amazon EventBridge

Thank you!

David Boyne
@boyney123



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